

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

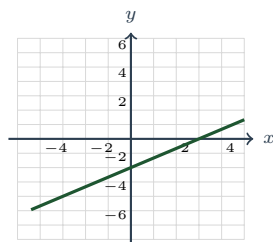
Write the letter of the best description next to each term.

- | | |
|--------------------------------|--|
| 1. Counterexample C | (A) The number $b^2 - 4ac$, which counts a quadratic's real solutions <i>before</i> you solve it. |
| 2. Closure D | (B) An equation true for every real number, so its solution set is all real numbers. |
| 3. Factor completely F | (C) One example that proves a general claim false; no number of successes can replace it. |
| 4. Literal equation E | (D) The property that performing an operation on members of a set always lands you back inside that set. |
| 5. Identity B | (E) An equation with more than one letter, solved for one letter in terms of the others. |
| 6. Discriminant A | (F) To rewrite an expression as a product — GCF first — until no factor breaks down further over the integers. |
| 7. Zero of a function H | (G) The constant ratio of change in output to change in input for a linear function; the slope. |
| 8. Rate of change G | (H) An input whose output is 0; also a solution of $f(x) = 0$ and an x -intercept of the graph. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Which property justifies the step $5(x - 3) = 5x - 15$?
 (A) Commutative property of multiplication (B) **Distributive property** ←
 (C) Associative property of multiplication (D) Multiplicative identity
- Simplified with no negative exponents, $\frac{18a^5b^{-3}}{6a^2b}$ is (A) $\frac{3a^3}{b^4}$ ← (B) $\frac{3a^3}{b^2}$ (C) $3a^3b^4$ (D) $\frac{12a^3}{b^4}$
- The solution of $-4x + 9 \geq 25$ is (A) $x \geq -4$ (B) $x \leq -4$ ← (C) $x \leq 4$ (D) $x \geq 4$
- How many *real* solutions does $2x^2 - 3x + 5 = 0$ have? (A) two (B) one (C) **none** ← (D) infinitely many
- Which relation is *not* a function?
 (A) $\{(-3, 1), (0, 1), (2, 1), (5, 1)\}$ (B) $\{(1, -3), (1, 0), (2, 5)\}$ ←
 (C) $\{(-2, 4), (-1, 1), (0, 0), (1, 1)\}$ (D) $y = 3x - 7$
- SOL-style.** Which equation represents the line shown?



(A) $y = \frac{3}{2}x - 2$

(B) $y = \frac{2}{3}x - 2$ ←

(C) $y = -\frac{2}{3}x - 2$

(D) $y = \frac{2}{3}x + 2$

Teacher Note

Item 1: the distributive property is the only one that *changes the number of terms* — that is the tell students should carry out of Lesson 1.0. Item 2: subtract exponents on division; $b^{-3-1} = b^{-4}$, then move it down. (B) subtracts as $-3 - 1 = -2$, (C) drops the negative on the way down, (D) subtracts the coefficients instead of dividing them. Item 3: dividing by -4 **flips** the inequality — (A) is the missed flip and (C)/(D) lose the sign of the endpoint. Item 4: $b^2 - 4ac = 9 - 40 = -31 < 0$. A student who tries to factor here has missed the point of the discriminant. Item 5: one input, one output — $x = 1$ is paired with both -3 and 0 . (A) repeats an *output*, which is allowed, and is the distractor worth sorting for. Item 6: from $(0, -2)$ the line goes up 2, right 3. (A) is rise and run swapped, (C) misreads the direction, (D) the sign of the intercept.

Part C — Short Answer & Computation (40 pts)

Show your work. A wrong answer with a right idea can still earn credit.

1. Numbers and justification.

(a) Label each number **R** (rational) or **I** (irrational).

$\sqrt{49}$ **R** $\sqrt{7}$ **I** $-\frac{5}{8}$ **R** $0.\overline{3}$ **R** π **I**

(b) Name the property that justifies each step.

$2(x + 5) = 2x + 10$ **distributive property**

$2x + 10 - 10 = 14 - 10$ **subtraction property of equality**

$3 + x = x + 3$ **commutative property of addition**

(c) “The set of integers is closed under division.” Disprove it.

One counterexample is enough: $3 \div 2 = 1.5$, and 1.5 is not an integer. Listing cases that *do* work ($6 \div 3 = 2$) proves nothing.

2. Translate and evaluate.

(a) Write an expression for “seven less than four times a number n .” $4n - 7$

(b) Evaluate $\frac{|a-b|}{2} + \sqrt{c}$ for $a = -3$, $b = 5$, $c = 16$. $a - b = -3 - 5 = -8$, so $|-8| = 8$ and $\frac{8}{2} = 4$; $\sqrt{16} = 4$.

Total: $4 + 4 = 8$. Take the absolute value *after* subtracting — the bars are grouping symbols.

(c) Evaluate $\sqrt[3]{x} - 2y$ for $x = -27$, $y = \frac{1}{2}$. $\sqrt[3]{-27} = -3$ (an *odd* index accepts a negative radicand), and $2 \cdot \frac{1}{2} = 1$, so $-3 - 1 = -4$.

3. Exponents and polynomial operations.

(a) $(3x^2y^4)(-2x^5y)$ (b) $\frac{20p^6q^2}{5p^6q^5}$ (no negative exponents)

(a) Multiply coefficients, *add* exponents: $-6x^7y^5$.

(b) Divide coefficients, *subtract* exponents: $4p^0q^{-3} = \frac{4}{q^3}$. The p 's cancel completely — $p^0 = 1$, not 0.

(c) $(2x - 5)(3x + 4)$ (d) $(4x^2 - x + 6) - (x^2 + 3x - 2)$

(c) $6x^2 + 8x - 15x - 20 = 6x^2 - 7x - 20$.

(d) The minus sign is a factor of -1 — it flips every sign in the second polynomial: $4x^2 - x + 6 - x^2 - 3x + 2 = 3x^2 - 4x + 8$.

4. **Factor completely.** GCF first, every time.

(a) $3x^2 - 27$ (b) $x^2 - 2x - 3$

(a) **GCF 3 first:** $3(x^2 - 9) = 3(x - 3)(x + 3)$. **Stopping at $3(x^2 - 9)$ is unfinished — the GCF *uncovered* the difference of squares.**

(b) **Product -3 , sum -2 : -3 and $+1$, so $(x - 3)(x + 1)$.**

(c) $2x^2 + 11x + 12$ (d) $4x^3 - 12x^2 + 8x$

(c) $a \cdot c = 24$; 3 and 8 add to 11: $2x^2 + 3x + 8x + 12 = x(2x + 3) + 4(2x + 3) = (2x + 3)(x + 4)$.

(d) **GCF $4x$:** $4x(x^2 - 3x + 2) = 4x(x - 1)(x - 2)$. **Dropping the x from the GCF loses a factor — and later, a solution.**

5. **Radicals and rational exponents.**

(a) $\sqrt{50} = 5\sqrt{2}$ (b) $\sqrt[3]{40} = 2\sqrt[3]{5}$

(c) Write $\sqrt{x^3}$ with a rational exponent. $x^{3/2}$

(d) Write $8^{1/3}$ as a radical, then evaluate it. $\sqrt[3]{8} = 2$

(e) $\sqrt{12} + \sqrt{27} =$ **Simplify first, decide second:** $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$. **Before simplifying these look like unlike radicals; $\sqrt{39}$ is the error to watch for.**

6. **Linear equations and inequalities.**

(a) Solve $3(2x - 4) = 4x + 2$. **Distribute:** $6x - 12 = 4x + 2$. **Subtract $4x$:** $2x - 12 = 2$. **Add 12:** $2x = 14$. **Divide by 2:** $x = 7$. **Check:** $3(14 - 4) = 30$ and $4(7) + 2 = 30$ ✓

(b) Classify each solution set as **one solution**, **no solution**, or **infinitely many**. You should not need to finish solving.

$5x + 3 = 5x - 8$ **no solution** $2(x + 6) = 2x + 12$ **infinitely many**

(c) Solve $A = \frac{1}{2}bh$ for h . $h = \frac{2A}{b}$

(d) Solve $-2x + 1 < 9$, then graph the solution on the number line.

$x > -4$



7. **Quadratic equations.**

(a) Solve by factoring: $x^2 - 2x - 3 = 0$ $(x - 3)(x + 1) = 0$, so by the zero product property $x = 3$ or $x = -1$. **The equation must be set equal to 0 *before* factoring — that is what makes the property apply.**

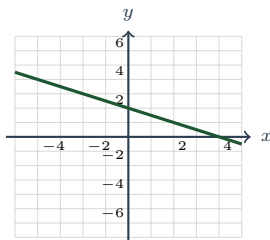
(b) Solve by square roots: $(x + 1)^2 = 49$ **Take the square root of both sides and write the $\pm$ yourself:** $x + 1 = \pm 7$, so $x = 6$ or $x = -8$. **Dropping the $\pm$ loses half the answer.**

(c) Solve with the quadratic formula: $x^2 + 4x - 6 = 0$ (exact answer) $a = 1$, $b = 4$, $c = -6$: $x = \frac{-4 \pm \sqrt{16 + 24}}{2} = \frac{-4 \pm \sqrt{40}}{2} = \frac{-4 \pm 2\sqrt{10}}{2} = -2 \pm \sqrt{10}$. **Simplify the radical, then divide *every* term by 2.**

(d) *Without solving*, use the discriminant to say how many real solutions $9x^2 - 12x + 4 = 0$ has. **one (a repeated real solution)**

Discriminant: $144 - 144 = 0$

8. **Read the graph.** The linear function g is graphed below; each gridline is one unit.



- (a) $g(-2) = 3$ (b) Solve $g(x) = 0$: $x = 4$
- (c) That number in (b) is called the **zero** of g , and on the graph it is the point **(4, 0)**.
- (d) Slope: $-\frac{1}{2}$; y -intercept: **(0, 2)**; equation of g : $y = -\frac{1}{2}x + 2$
- (e) Is g increasing or decreasing? **decreasing** Domain: **all real numbers**
- (f) Write the equation of the line through $(0, -3)$ **parallel** to g $y = -\frac{1}{2}x - 3$, and the line through $(0, 2)$ **perpendicular** to g $y = 2x + 2$.

Teacher Note

Item 1(c) is the Lesson 1.0 idea being assessed: a counterexample is a *single* number, and a list of successes earns nothing. Item 2(b): watch for $|-3| - |5|$ — the bars group the whole subtraction. Item 3(d) is the most-missed on this test; a student who writes $3x^2 + 2x + 4$ has distributed the minus to only the first term. Item 4: “completely” is the graded word — (a) without the second factoring step and (d) without the x in the GCF are both half answers. Item 6(b) should be answered by *looking*, not solving; if a student solved and got $3 = -8$, give full credit but note the shortcut. Item 6(d): the flip and the open circle are separate points — $\geq$ vs. $>$ is the endpoint question, and a filled dot here is wrong. Item 7(b): $x = 6$ only (no $\pm$) is the classic half-answer. Item 8(c) is the unit’s punchline — *zero*, *solution of $g(x) = 0$* , and *x -intercept* are three names for one thing; accept any correct naming but insist the point be an ordered pair. Item 8(f): the perpendicular slope is the negative *reciprocal*, so 2, not -2 and not $\frac{1}{2}$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Three families, six questions.** Each table below is generated by a linear, a quadratic, or an exponential rule.

x	$A(x)$	x	$B(x)$	x	$C(x)$
0	5	0	3	0	1
1	8	1	6	1	2
2	11	2	12	2	5
3	14	3	24	3	10

(a) Name the family of each table and give the numerical evidence — constant difference, constant *second* difference, or constant ratio. (b) Write a rule for each function. (c) Which function is largest at $x = 1$? At $x = 5$? Show the values. (d) A classmate says “ B is just the biggest one.” Use your answers to (c) to explain why “which is bigger” is not a fair question until someone says *where*. (e) If B modeled a population of bacteria in hours, what domain would make sense, and why is that narrower than the domain of the rule by itself? (a) **A is linear: the first differences are 3, 3, 3 — a constant difference. B is exponential: the ratios are $6/3 = 12/6 = 24/12 = 2$ — a constant ratio. C is quadratic: the first differences 1, 3, 5 are not constant, but their differences are 2, 2 — a constant *second* difference.**

(b) $A(x) = 3x + 5$ (slope 3, starts at 5); $B(x) = 3 \cdot 2^x$ (starting value 3, ratio 2); $C(x) = x^2 + 1$.

(c) At $x = 1$: $A(1) = 8$, $B(1) = 6$, $C(1) = 2$, so A is largest. At $x = 5$: $A(5) = 20$, $B(5) = 3 \cdot 32 = 96$, $C(5) = 26$, so B is largest — by a lot.

(d) The three functions trade places. At $x = 1$ the linear function is on top; by $x = 5$ the exponential has passed both of the others and is still pulling away. A constant ratio eventually beats a constant difference and even a constant second difference, but “eventually” is the whole

point — before the crossing, B is the *smallest* of the three. So the question “which is bigger” has no answer until the input is named.

(e) Hours cannot be negative, so the domain in context is $x \geq 0$ — and if the count is only taken on the hour, the whole numbers $0, 1, 2, 3, \dots$. The rule $3 \cdot 2^x$ by itself accepts every real number, including negatives. The context, not the algebra, is what shrinks the domain.

2. Modeling and solution sets. Two climbing gyms price a day pass differently.

Summit: a one-time \$40 membership fee, then \$12 per visit.

Basecamp: no fee, \$22 per visit.

(a) Write a function for the total cost at each gym after v visits. (b) Find the number of visits at which the two costs are equal. Show every step, and name the property of equality that justifies each one. (c) Verify your answer by substituting it into *both* functions. (d) Which gym is cheaper for 2 visits? For 10 visits? Give the costs. (e) Suppose Basecamp changed its pricing to a \$40 fee plus \$12 per visit. Setting the two costs equal would give $12v + 40 = 12v + 40$. What is the solution set, what is that kind of equation called, and what does it mean about the two gyms? (f) What domain makes sense for v in this situation, and why? (a) $S(v) = 12v + 40$ and $B(v) = 22v$. The fee is the y -intercept (a one-time cost); the per-visit price is the slope.

(b) $12v + 40 = 22v$. Subtract $12v$ from both sides (subtraction property of equality): $40 = 10v$. Divide both sides by 10 (division property of equality): $v = 4$ visits.

(c) $S(4) = 12(4) + 40 = 88$ and $B(4) = 22(4) = 88$. Both give \$88, so $v = 4$ checks. ✓

(d) At 2 visits: $S(2) = 64$, $B(2) = 44$ — Basecamp is cheaper, because the \$40 fee has not paid for itself yet. At 10 visits: $S(10) = 160$, $B(10) = 220$ — Summit is cheaper, by \$60. The break-even point found in (b) is exactly where the answer switches.

(e) Every v makes it true, so the solution set is all real numbers and the equation is an identity — the two sides are the same expression written twice. In context it means the gyms would charge identically no matter how many times you went, so cost could never decide between them.

(f) v counts visits, so it must be a whole number $0, 1, 2, 3, \dots$ — there is no such thing as 2.7 visits or -3 visits. The functions themselves would happily accept any real number; the situation is what restricts the domain.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) all three families named *with* the numerical evidence — a family named with no evidence earns half, since the evidence is the skill; 1 pt (b) all three rules ($3 \cdot 2^x$, not $2^x + 3$ — the starting value multiplies); 1 pt (c) both comparisons with values shown; 1 pt (d) the trade-places argument; 1 pt (e) $x \geq 0$ (or whole numbers); 1 pt (e) the contrast between the rule’s domain and the context’s. Do not accept “exponential always wins” in (d) — at $x = 1$ it loses, and that is the item. D2: 1 pt (a) both functions; 1 pt (b) $v = 4$; 1 pt (b) *both* properties named by name — this is the A.EI.1 justification clause and the reason Lesson 1.0 exists; 1 pt (c) the substitution into both; 1 pt (d) both comparisons with costs; 1 pt (e) identity *and* all real numbers *and* the interpretation. Award (f) inside the (e) point if (e) is otherwise complete. A student who answers (e) with “ $v = 0$ ” has confused an identity with a solved equation — reteach against the one/none/ininitely-many table from Lesson 1.2.